ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) | 979-8-3315-6701-9/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICASSP55912.2026.11464908

LLM-GUIDED SAM FOR LEFT VENTRICLE SEGMENTATION AND FUNCTIONAL ANALYSIS IN ECHOCARDIOGRAPHY

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ABSTRACT

Accurate segmentation of the left ventricle (LV) in echocardiography is crucial for cardiac structure and function assessment, yet manual annotation is time-consuming, subjective, and hard to standardize. The Segment Anything Model (SAM) provides a foundation for general segmentation, but its direct application to ultrasound is hindered by low contrast, blurred boundaries, and the absence of domain-specific knowledge. We propose **LV-SAM++**, an LLM-guided SAM-based framework for fully automated LV segmentation and functional assessment. LV-SAM++ introduces three key components: (1) LLM-driven semantic prompt generation to eliminate manual input and inject anatomical knowledge, (2) semantic-guided attention to align prompts with visual features for anatomically coherent segmentation, and (3) a locally enhanced CNN branch to refine fine-grained cardiac structures. A dedicated module further computes functional indices such as eccentricity index (EI) and area change ratio (ACR) directly from segmentation masks. On the CAMUS dataset, LV-SAM++ achieves a Dice score of 93.97%, a Hausdorff distance of 9.14 px, and state-of-the-art accuracy in EI and ACR, enabling reliable and clinically meaningful LV analysis in echocardiography.

***Index Terms*—** Left Ventricle Segmentation, Segment Anything Model (SAM), Functional Assessment, Large Language Model (LLM), Echocardiography

1. INTRODUCTION

Accurate segmentation of the left ventricle (LV) in echocardiography is essential for evaluating cardiac structure and systolic function, aiding early diagnosis and treatment of cardiovascular diseases [1]. Functional indices such as the eccentricity index (EI) and area change ratio (ACR) are widely used [2], yet current workflows still rely on manual contour delineation, which is time-consuming, subjective, and hard to standardize [3].

Traditional segmentation methods such as thresholding, region growing, and active contours [4] perform poorly under ultrasound noise and low contrast. Deep learning models like U-Net [5] and its variants [6, 7], as well as Transformer-based models such as TransUNet [8] and Swin-Unet [9], improve accuracy and global context modeling. Shape priors and attention mechanisms further enhance robustness against blurred boundaries [10]. However, these task-specific models still generalize poorly across patients, imaging protocols, and institutions, limiting clinical adoption.

Foundation models such as SAM [11], along with adaptations like MedSAM [12] and SAMed [13], introduced prompt-driven universal segmentation but remain limited for echocardiography due to reliance on manual prompts, lack of cardiac anatomical awareness, and poor performance under speckle noise [14]. Ferreira et al. [15] reported that SAM achieved only $r = 0.39$ correlation with clinical measurements and required over four hours of prompting for 100 images, highlighting its inefficiency and limited utility.

Several extensions have been proposed to mitigate these limitations. SAMUS [16] and MSA [17] introduced CNN branches or adapters to enhance local feature modeling, while AutoSAMUS [16] explored end-to-end automatic prompt generation. High-quality prompt designs such as point- or shape-constrained heatmaps [18] improve boundary localization, and Grounded-SAM [19] leverages textual descriptions to reduce manual intervention. Multimodal approaches further highlight the value of language guidance: CoOp [20], ALBEF [21], and BLIP-2 [22] align vision and text representations, while PubMedBERT [23] encodes domain-specific anatomical semantics. Such cross-modal integration is particularly useful for handling view-related anatomical variability

This work was supported by the National Natural Science Foundation of China (Grant Nos. 62576222 and 62266039), the Key Project of Guangdong Province General Higher Education Institutions in Specific Research Areas (Grant No. 2023ZDZX2055), and the Open Research Fund from the Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ) (Grant No. GML-KF-24-06). Additional support was provided in part by the Chinese Scholarship Council (CSC).

(e.g., apical two-chamber views) and improving segmentation robustness. However, existing methods still face three key limitations: (i) generated prompts often lack fine-grained anatomical semantics, (ii) structural priors remain insufficient under domain shift, and (iii) most pipelines focus solely on segmentation without integrating clinically meaningful functional validation.

To address these challenges, we propose **LV-SAM++**, a SAM-based framework for LV segmentation and functional assessment. It integrates LLM-guided semantic prompting, local feature enhancement, and prompt-aware feature weighting to improve boundary precision and robustness. EI and ACR are directly computed from segmentation outputs to validate clinical utility. The main contributions are:

- Semantic-Guided Prompt Learning: PubMedBERT-guided prompts with semantic-aware weighting embed anatomical knowledge and eliminate manual prompting.
- Local Feature Enhancement: A lightweight CNN branch with dilated convolutions captures clear LV boundaries under noisy and low-contrast conditions.
- Integrated Functional Analysis: EI and ACR are directly computed from segmentation masks to enable clinically meaningful functional validation.

2. METHOD

2.1. Overview

We propose **LV-SAM++**, an integrated framework for automated LV segmentation and functional assessment in echocardiography (Fig. 1). It inherits the SAM backbone but introduces three targeted modules: (1) a Local Feature Enhancement (LFE) branch, (2) a Semantic Prompt Learning (SPL) branch with Prompt-aware Feature Weighting (PaFW), and (3) a functional assessment module for deriving indices such as eccentricity index (EI) and area change ratio (ACR).

2.2. Local Feature Enhancement (LFE)

To compensate for ViT's limited local modeling, the LFE branch applies standard and dilated convolutions, expanding the receptive field without increasing parameters. The receptive field of a dilated convolution is:

$$\mathrm{RF}_{\text{dilated}} = (k-1) \times d + 1, \tag{1}$$

where k is kernel size and d is dilation rate. This improves sensitivity to weak LV boundaries in noisy ultrasound.

2.3. ViT Adaptation and PaFW

The SAM ViT encoder is adapted by reducing resolution, using overlapping patches, and inserting lightweight adapters.

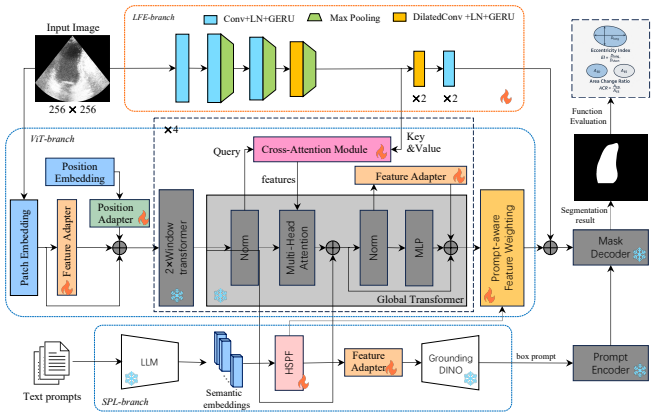


Fig. 1: Overview of **LV-SAM++**: LFE branch enhances boundary details, SPL branch integrates semantic prompts with PaFW, and the functional module computes EI and ACR.

To combine global ViT features with local CNN features, cross-branch attention is used. PaFW further strengthens semantic coupling by reweighting features $\mathbf{f}_i$ with respect to prompt embedding $\mathbf{t}$:

$$w_i = \frac{\exp(\cos(\mathbf{f}_i, \mathbf{t}))}{\sum_j \exp(\cos(\mathbf{f}_j, \mathbf{t}))}, \quad \mathbf{f}_i' = w_i \cdot \mathbf{f}_i, \tag{2}$$

where $\cos(\cdot)$ denotes cosine similarity. An entropy regularization encourages sparse, focused attention.

2.4. Structured Prompt Learning (SPL)

SPL introduces high-level anatomical priors via hierarchical medical prompts. Text descriptions are encoded with PubMedBERT and fused into a compact embedding, which is used both for PaFW reweighting and for bounding-box generation to guide SAM prompts. This replaces manual inputs with automated, semantically consistent prompts.

2.5. Functional Assessment

From predicted LV masks, two indices are computed: **Eccentricity Index (EI):**

$$EI = \sqrt{1 - \left(\frac{L_{\text{minor}}}{L_{\text{major}}}\right)^2}, \tag{3}$$

where $L_{\text{major}}, L_{\text{minor}}$ are ellipse axes. **Area Change Ratio (ACR):**

$$ACR = \frac{A_{ED} - A_{ES}}{A_{ED}}, \tag{4}$$

where A_{ED}, A_{ES} denote LV areas at end-diastole and end-systole. These extend segmentation into functional analysis.

2.6. Loss Function

The overall loss combines binary cross-entropy, Dice loss, and PaFW regularization:

$$\mathcal{L}_{total} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice} + \lambda_{\text{pfw}}\mathcal{L}_{\text{pfw}}, \tag{5}$$

with λ_{pfw} set to 0.1. Only CNN, adapters, and PaFW are trained, while the ViT backbone and language encoders are frozen for efficiency.

3. EXPERIMENT

3.1. Experimental Settings

3.1.1. Dataset

We evaluate LV-SAM++ on the CAMUS benchmark echocardiography dataset [10]. CAMUS contains 2D echocardiographic sequences from 500 patients with expert-annotated left ventricle (LV) contours in apical two- and four-chamber views, providing approximately 10,000 frames at a resolution of 512×512. The dataset presents several challenges, including blurred boundaries, speckle noise, and significant inter-patient shape variability. Following the official protocol, we focus on the apical two-chamber view, use the provided test set for evaluation, and reserve 10% of the training patients for validation.

3.1.2. Evaluation Metrics

Performance is assessed by structural accuracy and functional consistency. Structural accuracy is measured with Dice coefficient (overlap) and Hausdorff distance (boundary error). Functional consistency is evaluated via eccentricity index (EI) and area change ratio (ACR), computed from predicted masks, with accuracy quantified by mean absolute error (MAE) against clinical references [24, 25].

3.1.3. Implementation Details

LV-SAM++ is implemented in PyTorch and trained on an RTX 3090 GPU. Only LFE, CBA, HSPF, and adapters are updated, while other parameters remain frozen. Training uses Adam optimizer (lr $= 1 \times 10^{-2}$, weight decay 1×10^{-4}), StepLR decay every 50 epochs, batch size 16, and runs for 200 epochs. The checkpoint with the best validation Dice is reported.

3.2. Main Results

3.2.1. Quantitative Comparison

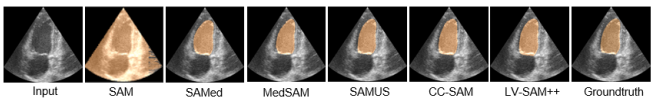
On the CAMUS dataset, LV-SAM++ consistently outperforms both foundation and task-specific models across structural (Dice, HD) and functional (EI, ACR) metrics (Table 1).

Table 1: Comparison of LV-SAM++ with foundation and task-specific models on the CAMUS dataset. Bold indicates the best results.

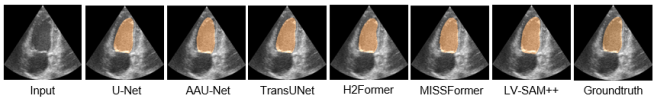
Method	Structural Metrics		Functional Metrics	
	Dice (%) ↑	HD (px) ↓	EI MAE ↓	ACR MAE ↓
Foundation models				
SAM [11]	28.18 ± 7.20	196.64 ± 35.11	–	–
MedSAM [12]	87.52 ± 4.90	15.28 ± 2.75	0.0239 ± 0.0031	0.091 ± 0.010
SAMed [13]	87.67 ± 4.80	13.24 ± 2.31	0.0242 ± 0.0034	0.094 ± 0.011
SAMUS [16]	91.13 ± 3.70	11.76 ± 1.95	0.0172 ± 0.0025	0.079 ± 0.009
CC-SAM [26]	92.85 ± 3.20	10.88 ± 1.80	0.0150 ± 0.0021	0.074 ± 0.008
Task-specific models				
U-Net [5]	93.56 ± 2.90	9.90 ± 1.62	0.0167 ± 0.0023	0.082 ± 0.008
SwinUNET [9]	91.88 ± 3.50	13.11 ± 2.12	0.0192 ± 0.0027	0.087 ± 0.009
SETR [27]	92.77 ± 3.10	11.50 ± 1.88	0.0167 ± 0.0024	0.082 ± 0.008
MISSFormer [28]	93.25 ± 2.95	9.94 ± 1.64	0.0163 ± 0.0021	0.084 ± 0.007
TransUNet [8]	93.60 ± 2.85	9.60 ± 1.55	0.0155 ± 0.0020	0.080 ± 0.007
AAU-Net [29]	93.59 ± 2.82	9.20 ± 1.48	0.0160 ± 0.0021	0.078 ± 0.006
H2Former [30]	93.44 ± 2.88	9.60 ± 1.51	0.0153 ± 0.0020	0.077 ± 0.006
LV-SAM++ (Ours)	**93.97 ± 2.65**	**9.14 ± 1.42**	**0.0142 ± 0.0018**	**0.071 ± 0.006**

Compared with representative SAM-based foundation models, LV-SAM++ yields markedly higher Dice scores and lower HD values, while also reducing errors in functional indices. The extremely low performance of vanilla SAM is mainly due to its reliance on manual prompts and lack of ultrasound-specific adaptation, which leads to severe failures under speckle noise and low-contrast conditions. Against CNN-, Transformer-, and hybrid-based task-specific architectures, LV-SAM++ further achieves superior segmentation accuracy and more accurate estimation of functional indices. These improvements translate into smoother, more anatomically coherent contours and more reliable estimation of clinically relevant parameters, demonstrating the effectiveness of LV-SAM++ for comprehensive LV assessment.

3.2.2. Qualitative Comparison



(a) **Foundation models.** LV-SAM++ generates more accurate LV boundaries on CAMUS.



(b) **Task-specific models.** LV-SAM++ shows more continuous contours and fewer errors.

Fig. 2: Qualitative comparison of LV-SAM++ against (a) foundation models and (b) task-specific models on the CAMUS dataset. LV-SAM++ produces visually more accurate and anatomically consistent LV boundaries.

Figure 2 illustrates a qualitative comparison between LV-SAM++ and representative (a) foundation models and (b)

task-specific networks on the CAMUS dataset. LV-SAM++ produces smoother and more anatomically consistent left ventricular boundaries, closely matching the ground truth even in challenging low-contrast cases, while baseline methods often exhibit boundary discontinuities or under-segmentation.

3.2.3. *Functional Evaluation*

We further assess the clinical utility of LV-SAM++ using two functional indices, the eccentricity index (EI) and the area change ratio (ACR), derived from predicted LV masks on the CAMUS dataset. As shown in Table 1, LV-SAM++ achieves the lowest MAE for EI (0.0142 ± 0.0018) and ACR (0.071 ± 0.006), surpassing both foundation and task-specific models. These results confirm that improved segmentation quality leads to more accurate and reliable frame-level cardiac functional estimation.

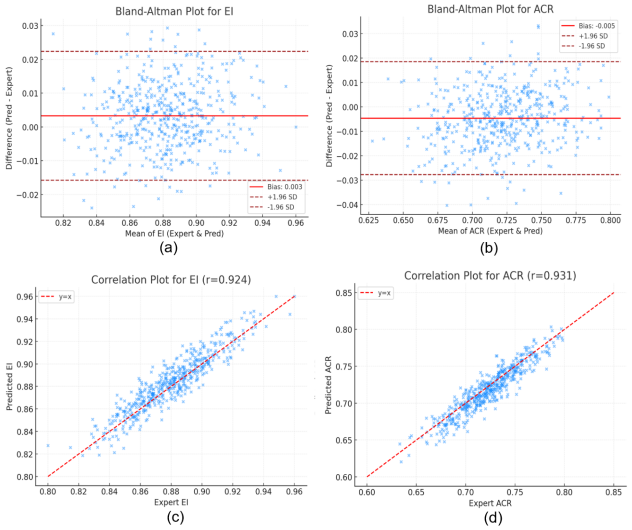


Fig. 3: Functional evaluation on CAMUS: (a,b) Bland–Altman plots for EI and ACR show small biases and narrow limits of agreement. (c,d) Correlation plots demonstrate strong agreement with expert measurements.

Fig. 3 shows that Bland–Altman plots reveal minimal bias (EI: $+0.003$, ACR: -0.005) and narrow 95% limits of agreement, while correlation plots indicate strong agreement with expert values (EI: $r = 0.924$, ACR: $r = 0.931$). These results confirm that LV-SAM++ delivers both accurate segmentation and clinically reliable functional assessment within a unified pipeline.

3.2.4. *Ablation Study*

We performed ablation experiments to evaluate the contribution of each module in LV-SAM++. Starting from the full model, we removed one component at a time under identical training settings. The tested components include the Prompt-aware Feature Weighting (PaFW) module, the Hierarchical Semantic Prompt Fusion (HSPF) submodule, the entire Structured Prompt Learning (SPL) branch, and the Local Feature Enhancement (LFE) branch.

Table 2: Ablation results of LV-SAM++ on CAMUS. Bold indicates the best.

Configuration	Dice (%) ↑	HD (px) ↓	EI MAE ↓	ACR MAE ↓
w/o PaFW	93.21	9.52	0.0150	0.078
w/o HSPF (SPL)	93.08	9.63	0.0154	0.079
w/o SPL branch	92.84	9.88	0.0162	0.081
w/o LFE	92.60	10.02	0.0167	0.083
Full LV-SAM++	**93.97**	**9.14**	**0.0142**	**0.071**

As shown in Table 2, removing any module reduces performance. The largest drop occurs without LFE, confirming its importance for delineating blurry boundaries. Excluding SPL or HSPF mainly affects functional metrics, showing that semantic prompts improve clinical parameter estimation. Overall, all modules complement each other, jointly enhancing segmentation accuracy and the accuracy of functional index estimation.

4. DISCUSSION

LV-SAM++ integrates automatic semantic prompt generation, prompt-guided attention, and local feature enhancement within a SAM-based framework to achieve accurate left ventricular segmentation and reliable functional index estimation in echocardiography. By eliminating manual prompting, the method improves reproducibility and effectively addresses low-contrast imaging and boundary ambiguity, achieving competitive or superior performance compared with task-specific networks without post-processing. Nevertheless, several limitations remain: functional indices are derived solely from endocardial masks without epicardial information, prompt generation relies on image-derived features without patient-specific clinical inputs, and evaluation is limited to a single public dataset (CAMUS), leaving generalizability to multi-center settings unverified. Future work will extend the framework to epicardial segmentation, incorporate patient-aware and multi-view semantic prompts, and validate the approach on larger and more diverse clinical cohorts to enhance robustness and applicability.

5. CONCLUSION

We proposed LV-SAM++, a SAM-based framework for automated left ventricle segmentation and functional analysis in echocardiography. By integrating automatic prompt generation and a locally enhanced CNN branch, LV-SAM++ improves boundary accuracy and the reliability of clinically relevant functional indices. Experiments on the CAMUS dataset show that LV-SAM++ consistently outperforms both foundation and task-specific models.

6. REFERENCES

[1] Serkan Kiranyaz, Aysen Degerli, Tahir Hamid, et al., "Left Ventricular Wall Motion Estimation by Active Polynomials for Acute Myocardial Infarction Detection," *IEEE Access*, vol. 8, pp. 210301–210317, 2020.

[2] Nicolas Graziano, Kris Gommeren, Annelies Valcke, et al., "Retrospective evaluation of left ventricular eccentricity index in the assessment of precapillary pulmonary hypertension in dogs (2017–2021): 145 cases," *Frontiers in Veterinary Science*, vol. 12, pp. 1548417, 2025.

[3] Xiangbin Liu, Liping Song, Shuai Liu, and Yudong Zhang, "A Review of Deep-Learning-Based Medical Image Segmentation Methods," *Sustainability*, vol. 13, no. 3, pp. 1224, 2021.

[4] Dzung L Pham, Chenyang Xu, and Jerry L Prince, "Current Methods in Medical Image Segmentation," *Annual Review of Biomedical Engineering*, vol. 2, no. 1, pp. 315–337, 2000.

[5] Olaf Ronneberger, Philipp Fischer, and Thomas Brox, "U-net: Convolutional networks for biomedical image segmentation," in *MICCAI*. Springer, 2015, pp. 234–241.

[6] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang, "UNet++: A Nested U-Net Architecture for Medical Image Segmentation," in *DLMIA*. Springer, 2018, pp. 3–11.

[7] Ozan Oktay, Jo Schlemper, Loic Le Folgoc, et al., "Attention u-net: Learning where to look for the pancreas," in *MIDL*, 2018.

[8] Jieneng Chen, Yongyi Lu, Qihang Yu, et al., "Transunet: Transformers make strong encoders for medical image segmentation," *arXiv preprint arXiv:2102.04306*, 2021.

[9] Hu Cao, Yueyue Wang, Joy Chen, et al., "Swin-unet: Unet-like pure transformer for medical image segmentation," in *ECCV*. Springer, 2022, pp. 205–218.

[10] Sarah Leclerc, Erik Smistad, Joao Pedrosa, et al., "Deep Learning for Segmentation Using an Open Large-Scale Dataset in 2D Echocardiography," *IEEE Transactions on Medical Imaging*, vol. 38, no. 9, pp. 2198–2210, 2019.

[11] Alexander Kirillov, Eric Mintun, Nikhila Ravi, et al., "Segment anything," in *ICCV*, 2023, pp. 4015–4026.

[12] Jun Ma, Yuting He, Feifei Li, et al., "Segment anything in medical images," *Nature Communications*, vol. 15, no. 1, pp. 654, 2024.

[13] Kaidong Zhang and Dong Liu, "Customized Segment Anything Model for Medical Image Segmentation," *arXiv preprint arXiv:2304.13785*, 2023.

[14] Yuhao Huang, Xin Yang, Lian Liu, et al., "Segment anything model for medical images?," *Medical Image Analysis*, vol. 92, pp. 103061, 2024.

[15] Danielle L Ferreira and Rima Arnaout, "Are AI Foundation Models Efficient for Segmentation of Echocardiograms?," *Journal of the American Society of Echocardiography*, 2025.

[16] Xian Lin, Yangyang Xiang, Li Yu, and Zengqiang Yan, "Beyond Adapting SAM: Towards End-to-End Ultrasound Image Segmentation via Auto Prompting," in *MICCAI*. Springer, 2024, pp. 24–34.

[17] Junde Wu, Ziyue Wang, Mingxuan Hong, et al., "Medical SAM adapter: Adapting segment anything model for medical image segmentation," *Medical Image Analysis*, vol. 102, pp. 103547, 2025.

[18] Chengyi Wen and Lianghua He, "PSAM: Prompt-based Segment Anything Model Adaption for Medical Image Segmentation," in *SMC*. IEEE, 2024, pp. 3324–3329.

[19] Tianhe Ren, Shilong Liu, Ailing Zeng, et al., "Grounded SAM: Assembling Open-World Models for Diverse Visual Tasks," *arXiv preprint arXiv:2401.14159*, 2024.

[20] Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu, "Learning to Prompt for Vision-Language Models," *International Journal of Computer Vision*, vol. 130, no. 9, pp. 2337–2348, 2022.

[21] Junnan Li, Ramprasaath Selvaraju, Akhilesh Gotmare, et al., "Align before Fuse: Vision and Language Representation Learning with Momentum Distillation," *NeurIPS*, vol. 34, pp. 9694–9705, 2021.

[22] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi, "BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models," in *ICML*. PMLR, 2023, pp. 19730–19742.

[23] Yu Gu, Robert Tinn, Hao Cheng, et al., "Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing," *ACM Transactions on Computing for Healthcare*, vol. 3, no. 1, pp. 1–23, 2021.

[24] Attila Nemes, Árpád Kormányos, Nóra Ambrus, and Csaba Lengyel, "Left ventricular deformation in the presence of left ventricular 'rigid body rotation'in healthy adults—three-dimensional speckle-tracking echocardiographic insights from the magyar-healthy study," *Quantitative Imaging in Medicine and Surgery*, vol. 14, no. 7, pp. 4605, 2024.

[25] Yuzo Yamasaki, Michinobu Nagao, Takeshi Kamitani, et al., "Clinical impact of left ventricular eccentricity index using cardiac MRI in assessment of right ventricular hemodynamics and myocardial fibrosis in congenital heart disease," *European Radiology*, vol. 26, no. 10, pp. 3617–3625, 2016.

[26] Shreyank N Gowda and David A Clifton, "CC-SAM: SAM with Cross-Feature Attention and Context for Ultrasound Image Segmentation," in *ECCV*. Springer, 2024, pp. 108–124.

[27] Sixiao Zheng, Jiachen Lu, Hengshuang Zhao, et al., "Rethinking Semantic Segmentation From a Sequence-to-Sequence Perspective With Transformers," in *CVPR*, 2021, pp. 6881–6890.

[28] Xiaohong Huang, Zhifang Deng, Dandan Li, and Xueguang Yuan, "MISSFormer: An Effective Medical Image Segmentation Transformer," *arXiv preprint arXiv:2109.07162*, 2021.

[29] Gongping Chen, Lei Li, Yu Dai, et al., "AAU-Net: An Adaptive Attention U-Net for Breast Lesions Segmentation in Ultrasound Images," *IEEE Transactions on Medical Imaging*, vol. 42, no. 5, pp. 1289–1300, 2022.

[30] Along He, Kai Wang, Tao Li, et al., "H2Former: An Efficient Hierarchical Hybrid Transformer for Medical Image Segmentation," *IEEE Transactions on Medical Imaging*, vol. 42, no. 9, pp. 2763–2775, 2023.